Time Budget of Feeding in Northern Shovelers

at Mueller Park, Austin, Texas

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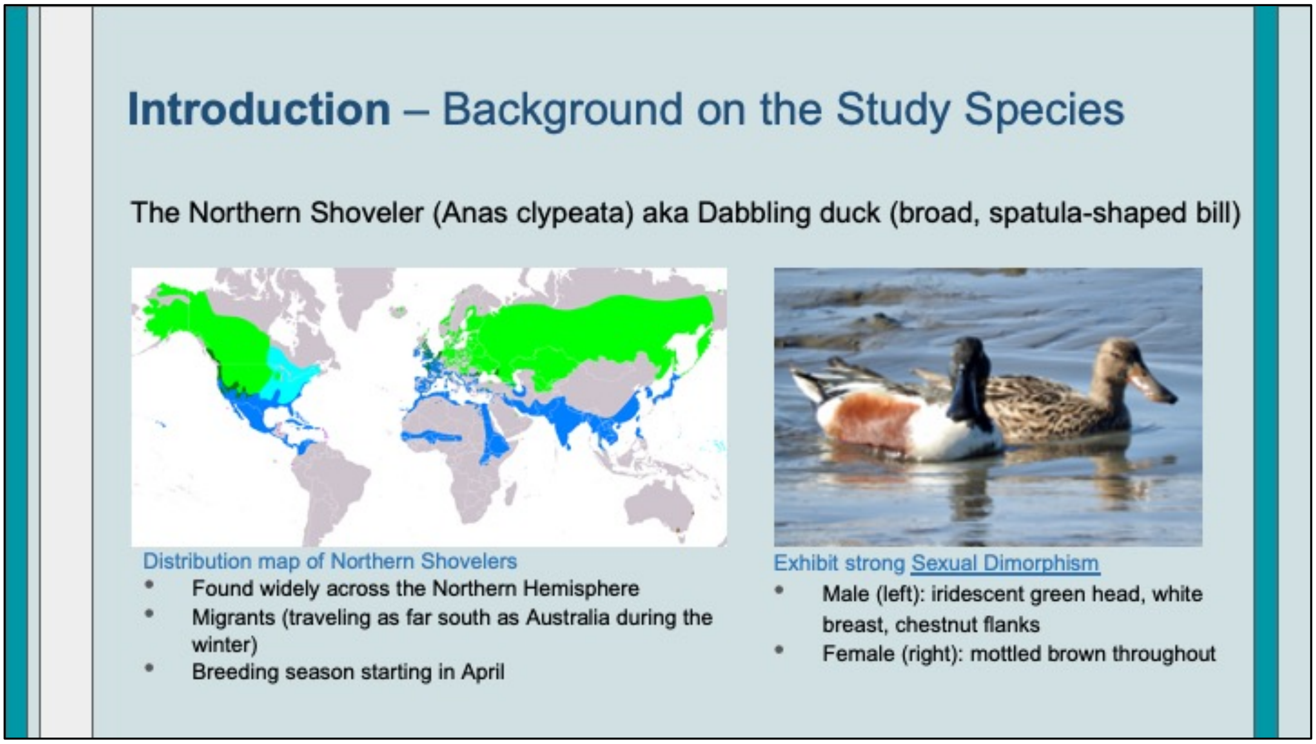
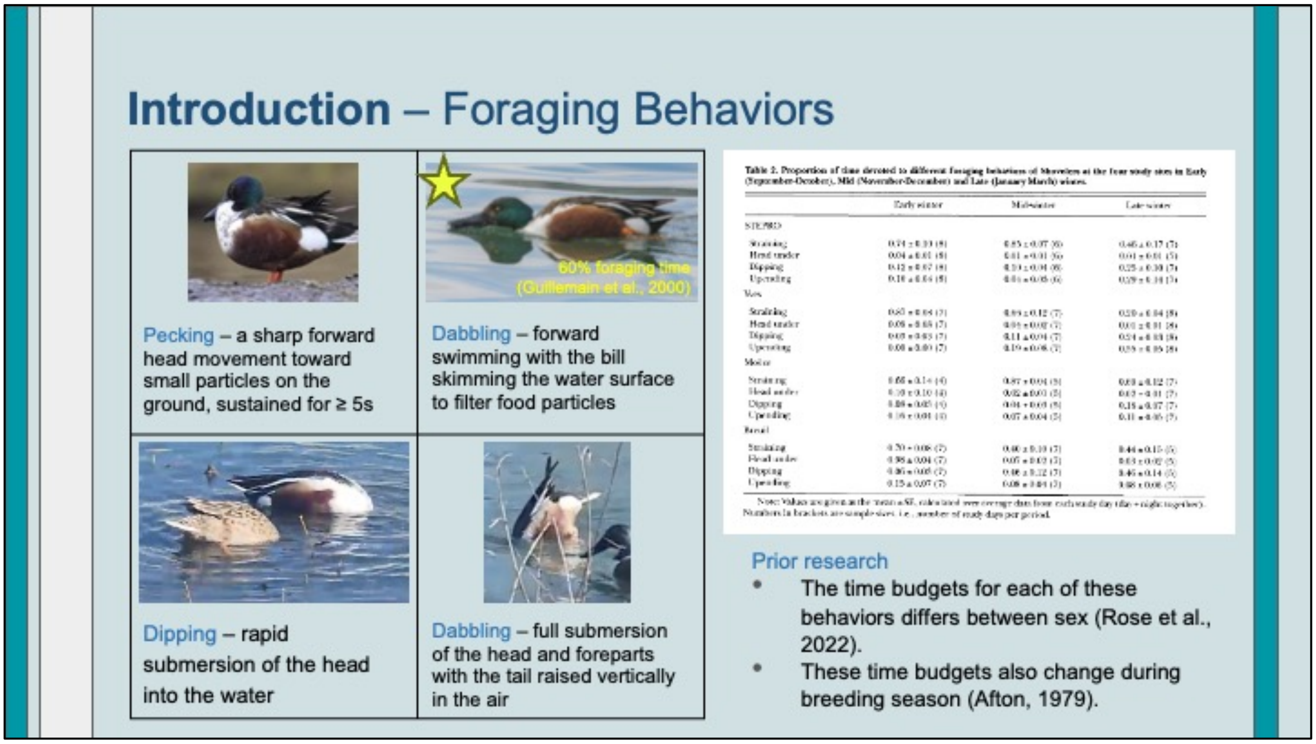
Discussion Section: Friday 9–10 AM

Hello everyone. My name is Kim Luong, and I'll be presenting our group's research project on the foraging behavior of Northern Shovelers observed at Mueller Lake Park here in Austin, Texas.
This project was conducted by Sergio Carranco, Kim Luong, and Peter Zhang.



So, who exactly is the Northern Shoveler? Scientifically known as Anas clypeata, the Northern Shoveler is a dabbling duck recognizable by its broad, spatula-shaped bill, which it uses to strain water for crustaceans, plankton, and small fish. You might recognize them from Mueller Park right here in Austin, where they're a fairly common sight. Widely distributed across the Northern Hemisphere, Northern Shovelers are highly migratory, with some populations wintering as far south as Australia. The species exhibits strong sexual dimorphism, which is when males and females look noticeably different. Males display a striking iridescent green head, white breast, and chestnut flanks, while females are mottled brown throughout. Their breeding season begins in April, shortly after the window of this study.

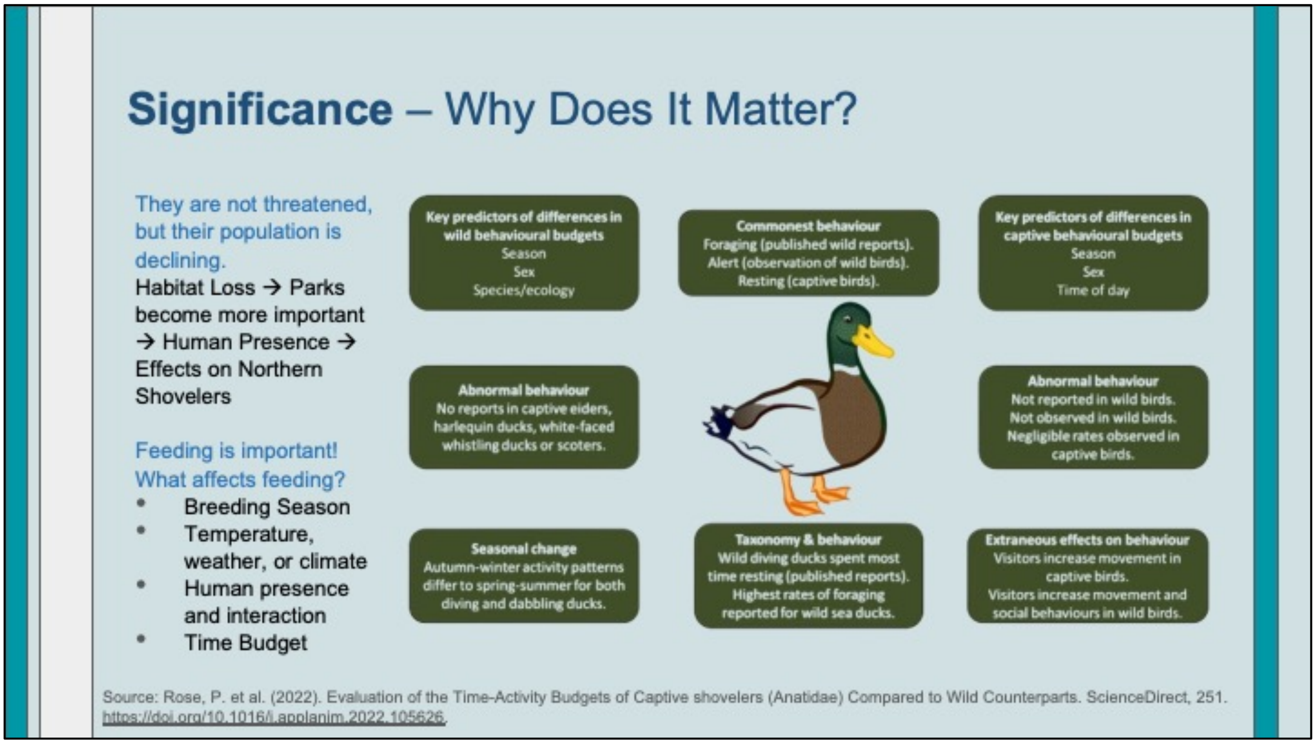


Northern Shovelers are unique feeders. They use their wide, spoon-shaped bills to strain water and capture crustaceans, plankton, and small fish.

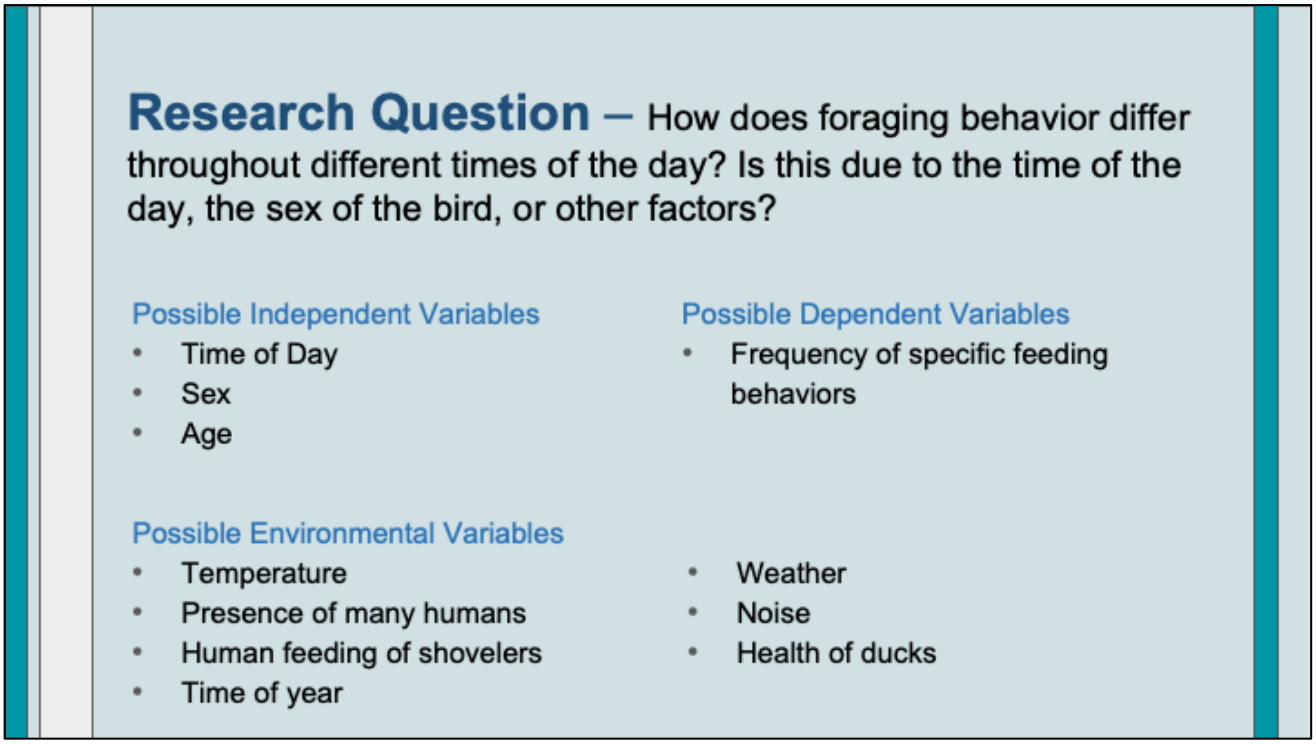
Northern Shovelers employ four primary foraging behaviors that were the focus of this study:

- Pecking — a sharp forward head movement toward small particles on the ground or water surface, sustained for at least 5 seconds.
- Dabbling — forward swimming with the bill skimming the water surface to filter food particles.
- Dipping — rapid submersion of the head into the water.
- Upending — full submersion of the head and foreparts with the tail raised vertically in the air.

Prior literature has established that the relative frequency of these behaviors varies between sexes and seasons. Afton (1979) found that females spent significantly more time foraging than males during the early breeding season, while Guillemain et al. (2000) documented that straining (analogous to dabbling in this study) represented the dominant foraging mode in wintering shovelers in France, accounting for over 60% of all foraging time.



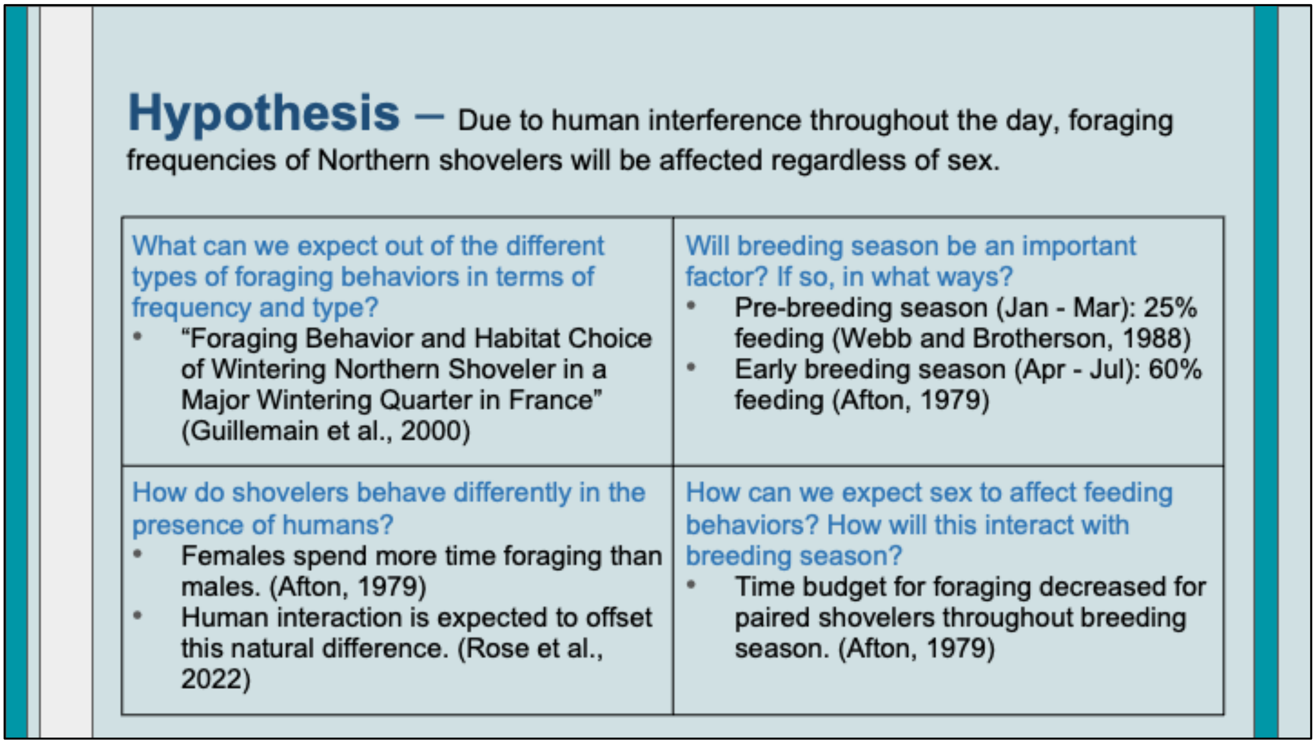
- Now, why is studying foraging behavior important? Northern Shovelers are not currently threatened, but their population is declining, largely due to habitat loss. As natural habitats shrink, parks like Mueller become increasingly critical refuges. But parks come with something natural habitats don't — a constant human presence. And that raises an important question: does human activity affect how shovelers feed?
- Feeding is fundamental to their survival and reproduction, so anything that disrupts their foraging efficiency has real biological consequences. Our study aimed to understand what's driving changes in foraging patterns — is it the time of day, the sex of the bird, or the human activity around them?



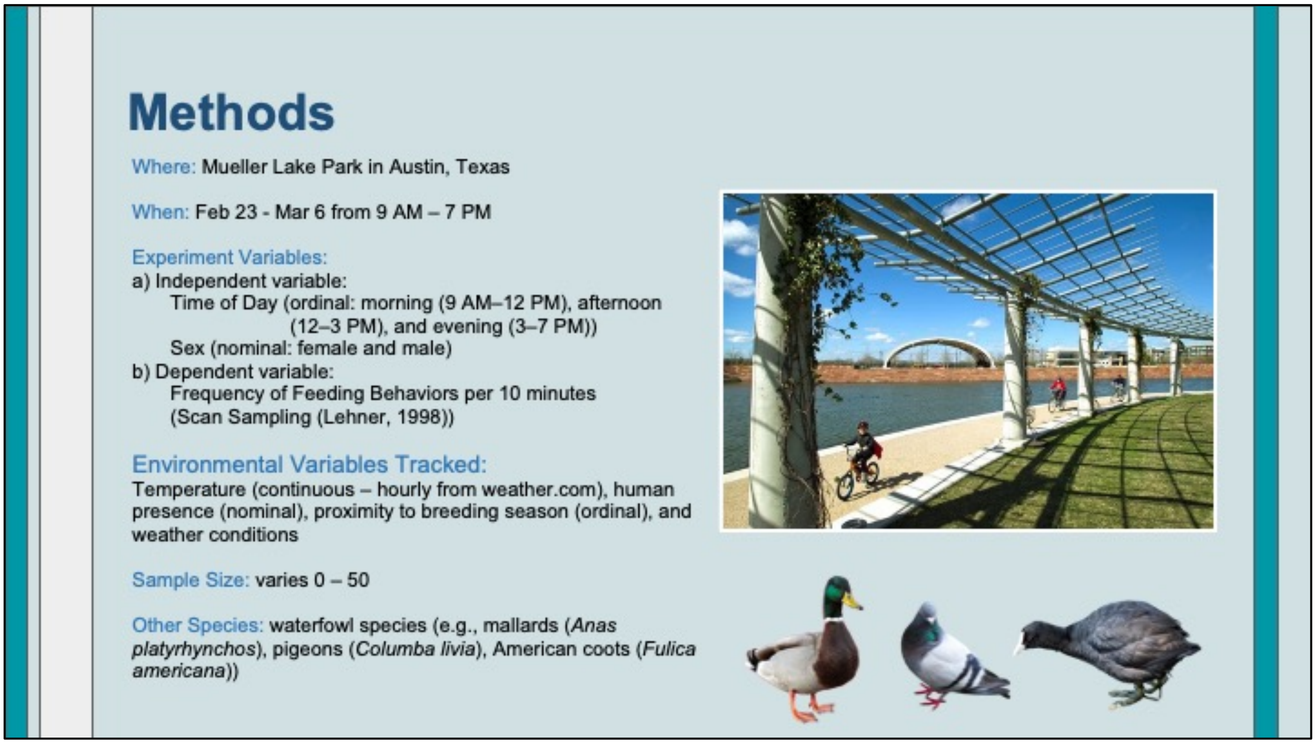
That brings us to our research question: How does foraging behavior differ throughout different times of the day? Is this due to the time of day, the sex of the bird, or other factors?

We identified our key variables as follows:

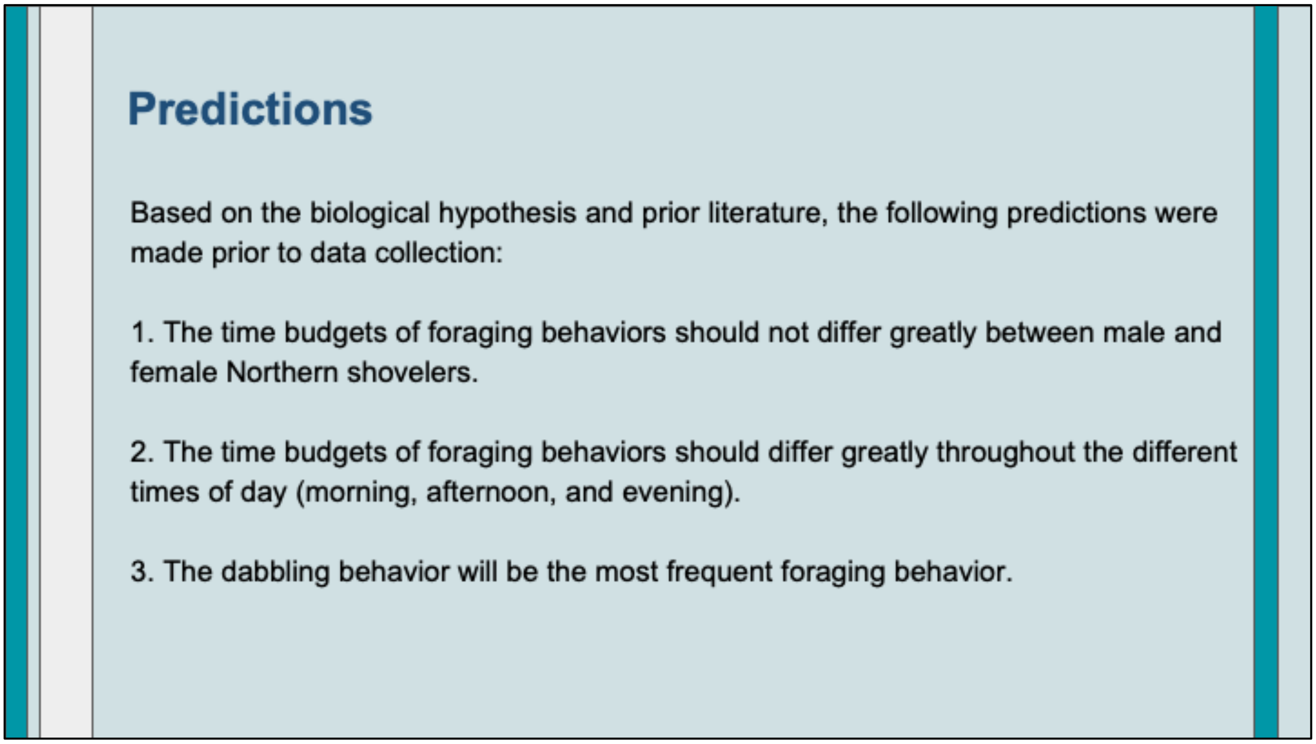
- Independent variables: Time of day (morning, afternoon, evening) and sex (male or female)
- Dependent variable: The frequency of each of the four foraging behaviors per 10-minute observation interval
- Environmental variables we tracked: Temperature, human presence, human feeding of the ducks, time of year, weather, and noise



- Our biological hypothesis was: **Due to human interference throughout the day, foraging frequencies of Northern Shovelers will be affected regardless of sex.**
- This hypothesis was motivated by two lines of evidence. First, Afton (1979) established that females naturally forage more than males, while males spend more time in alert postures. Second, Rose et al. (2022) demonstrated that human visitors reduce duck alertness in captive and semi-captive settings, such as a public park with frequent visitors. We reasoned this effect would offset the natural sex difference in foraging time.
- We also drew on documented foraging behavior of wintering Northern Shovelers in France, and Webb and Brotherson (1988), who reported that shovelers spent about 25% of their time feeding during pre-breeding months — a window that closely matches our February–March study period.
- Additionally, Guillemain et al. (2000) described daily variation in foraging location and behavior in wild shovelers, and we hypothesized that human activity throughout the day would modulate this variation in ways not observed in undisturbed populations.



- Observations were conducted at Mueller Lake Park (Austin, Texas) from February 23 to March 6. This site was selected because it supports a reliably visible Northern Shoveler population during the late winter period and provides open sightlines for scan sampling. Observations took place daily from 9:00 AM to 6:00 PM, with data collected across three time blocks: morning (9 AM–12 PM), afternoon (12–3 PM), and evening (3–7 PM).
- Scan sampling (Lehner, 1998) was used because all four target behaviors are state behaviors with clear, observable onset and offset points. This method enabled simultaneous recording of behavior across all visible individuals at a given moment, making it efficient for tracking population-level time budgets across large, variable group sizes. At 10-minute intervals, observers scanned all visible Northern Shovelers and recorded the count of individuals of each sex performing each of the four foraging behaviors. Six scan samples were collected per observation hour, and observers documented their presence via timestamped photographs at each site visit.
- Morning sessions were characterized by cooler temperatures (64–68°F) and lower numbers of park visitors. Afternoon and evening sessions were warmer (73–78°F) and coincided with higher human foot traffic.
- Shoveler sample sizes ranged dramatically, from zero to as many as 50 birds on a given day. While other waterfowl species — including mallards, American coots, and pigeons — were present at the site, no systematic effects of these species on shoveler behavior were observed.



Before analyzing our data, we made three key predictions:

- The time budgets of foraging behaviors would not differ significantly between male and female shovelers, as human presence in the park was expected to neutralize the naturally higher female foraging rate documented by Afton (1979)
- The time budgets of foraging behaviors would differ significantly across the three time periods — driven by human activity and the shovelers' own daily rhythms.
- **Dabbling would be the most frequent foraging behavior** — consistent with Guillemain et al. (2000), who found straining (what we call dabbling) represented over 60% of feeding behavior in wild shovelers.

Results – Data Distribution (Normality Summary)

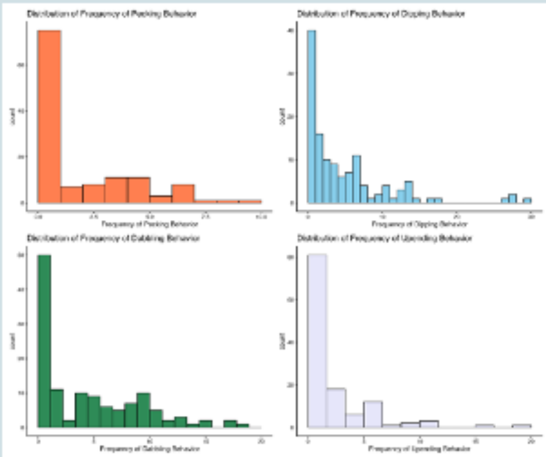


Figure 1. Histograms showing the distribution of frequency of the four different types of foraging behaviors.

Behavior	W	p-value
pecking	0.72116	3.66E-14
dabbling	0.7334	7.84E-14
dipping	0.86133	1.68E-09
upending	0.73759	1.17E-13

Table 1. Results of the Shapiro-Wilk's Normality Test

Trends:

- All histograms are skewed to the right.
- All p-values are less than 0.05.

- Before running our main statistical tests, we first checked whether our data followed a normal distribution using the Shapiro-Wilk test. The results are shown here in Table 1.
- All four behaviors — pecking, dabbling, dipping, and upending — had p-values far below 0.05, and the corresponding histograms in Figure 1 all show a strong right skew. This means the normality assumption was violated, which directed us toward non-parametric statistical tests for our main analyses.

Results – Sex Comparison (Summary Statistics)

	pecking		**dabbling**		**dipping**		**upending**	
Sex	**mean**	**sd**	**mean**	**sd**	**mean**	**sd**	**mean**	**sd**
male	0.8095238	1.342918	2.571429	3.0553	2.365079	3.158742	1.079365	1.998412
female	0.6746032	1.130164	1.793651	2.263864	2.007937	3.031821	0.8809524	1.562599
Wilcox Test:	**W** = 134036		**p-value** = 0.1					

Table 2. Summary Statistics of Frequency of Foraging Behaviors Between Sex

Trends:

- The mean of the frequency between male and female shovelers are similar for each of the foraging behaviors observed
- The standard deviation or the spread of the data are also similar for each of the foraging behaviors observed.
- The frequency and standard deviation of the dabbling and dipping behavior is greater than the frequency of the pecking and upending behavior.

Statistical Test - Wilcoxon-Mann Whitney test:

- 2 separate and independent groups
- normality assumption is violated.

- Now for the main results. First, we looked at whether foraging behavior differed between males and females.
- Table 2 summarizes the mean and standard deviation of each foraging behavior broken down by sex. You can see that the values are quite close between males and females across all four behaviors. The means for dabbling and dipping were higher than pecking and upending in both sexes.
- To test this statistically, we used the **Wilcoxon-Mann Whitney test**, which is appropriate for two independent non-normal groups. The result: **W = 134,036, p-value = 0.1**. Since $p > 0.05$, we fail to reject the null hypothesis — there is **no significant difference** in foraging frequency between males and females.

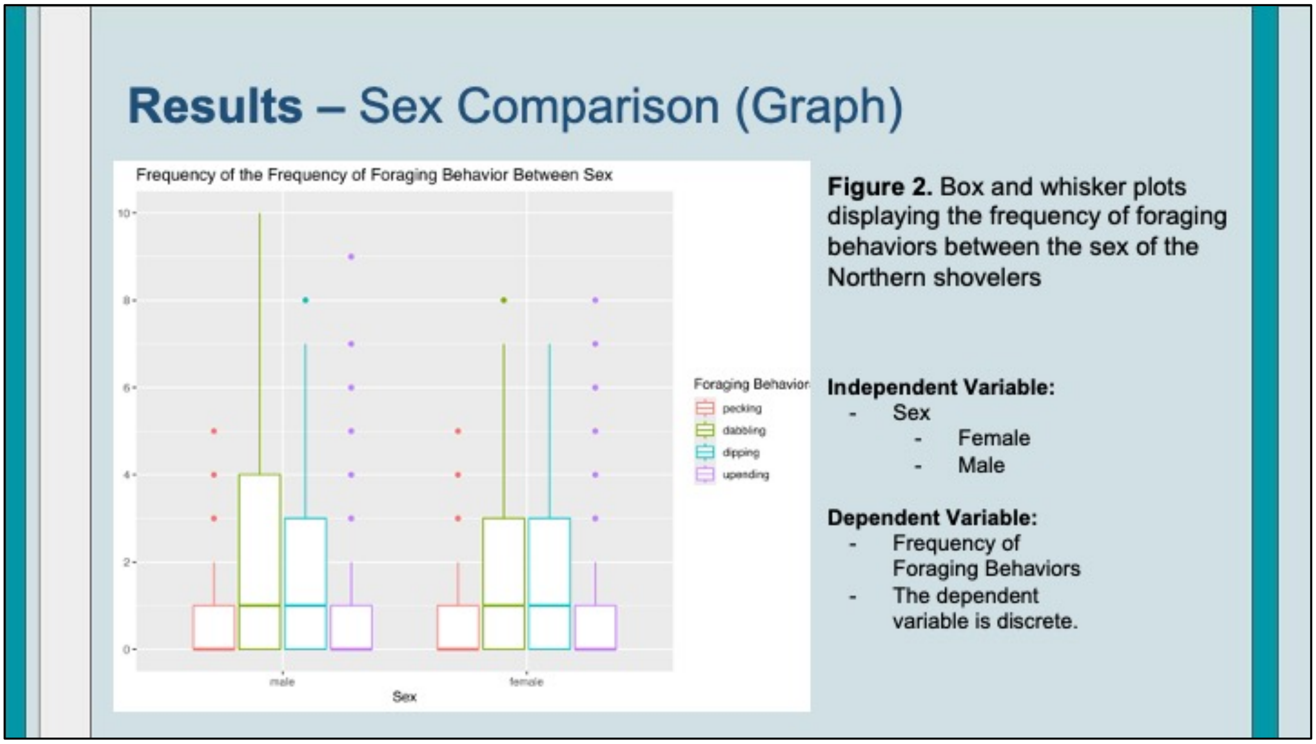


Figure 2 shows the box-and-whisker plots for foraging behavior frequency broken down by sex. As you can see visually, the distributions overlap considerably, and the medians are similar across all four behaviors for males and females. This is consistent with our statistical result — sex did not significantly predict foraging frequency.

Results – Time of Day Comparison (Summary Statistics)

	pecking		dabbling		dipping		upending	
Time of Day	**mean**	**sd**	**mean**	**sd**	**mean**	**sd**	**mean**	**sd**
morning	0.5952381	1.498935	2.404762	3.901591	1.142857	1.920003	0.6190476	1.188407
afternoon	0.8571429	1.675817	2.809524	3.307332	2.5	3.140996	1	1.861025
evening	3	2.419434	7.880952	4.484131	9.47619	7.249014	4.238095	4.326941
Kruskal Wallis Test:	**chi-squared** = 150.26		**df** = 2		**p-value** < 2.2e-16			

Table 3. Summary Statistics of Frequency of Foraging Behaviors Between Time of Day

Trends:

- The frequency of the four different types of foraging behaviors observed increases as the day progresses.
- The frequency of the dabbling and dipping behavior is greater than the frequency of the pecking and upending behavior.

Statistical Test - Kruskal-Wallis test:

- 3/+ independent groups
- normality assumption is violated.

- Next, we looked at how foraging behavior changed throughout the day.
- Table 3 shows a clear trend: the frequency of **all four foraging behaviors increased steadily from morning to evening**. For example, dabbling had a mean of about 2.4 in the morning, 2.8 in the afternoon, and 7.9 in the evening. The same pattern holds for dipping, pecking, and upending.
- To test this, we used the **Kruskal-Wallis test**, appropriate for three or more independent non-normal groups. The result: **chi-squared = 150.26, df = 2, p-value < 2.2e-16** — a highly significant result. The null hypothesis is rejected: foraging frequency **does differ significantly** by time of day.

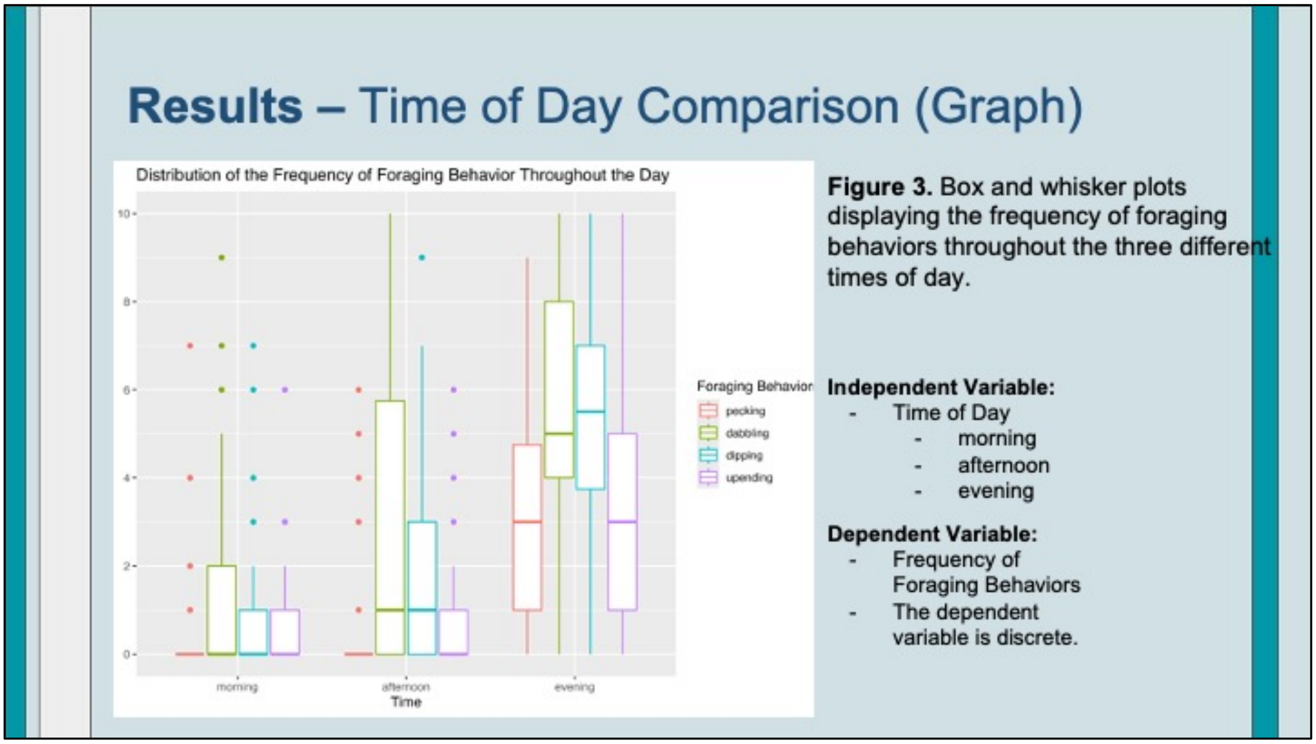


Figure 3 visualizes this clearly with box-and-whisker plots by time of day. Each of the four foraging behaviors shows a gradual increase from morning to afternoon to evening, with the evening showing notably higher medians and wider spreads. The statistical significance we found is plainly visible in these distributions.

Results – Paired Comparisons Between Behaviors

Data	t	df	p-value	CI	mean difference
pecking and dabbling	-8.1767	125	2.79E-13	-3.578269, -2.183636	-2.880952
pecking and dipping	-6.1191	125	1.12E-08	-3.823248, -1.954530	-2.888889
pecking and upending	-1.8644	125	0.06461	-0.98167582, 0.02929487	-0.4761905
dabbling and dipping	-0.016945	125	0.9865	-0.9348864, 0.9190134	-0.007936508
dabbing and upending	6.2277	125	6.60E-09	1.640548, 3.168976	2.404762
dipping and upending	5.0336	125	1.64E-06	1.464063, 3.361334	2.412698

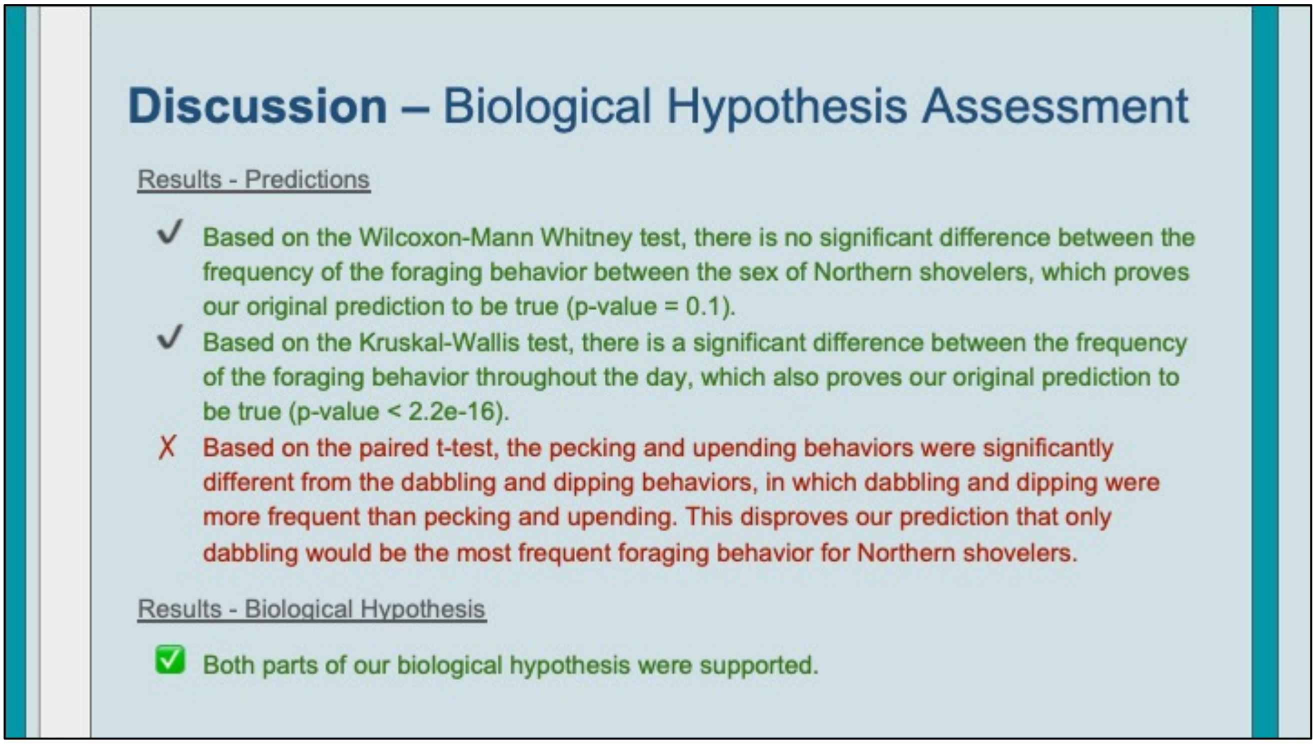
Table 4. Summary Statistics of Paired t-test Between Foraging Behaviors

Trends:

- There is a significant difference between the frequency of pecking and dabbling, the frequency of pecking and dipping, the frequency of dabbling and dipping, the frequency of dabbling and upending, and the frequency of dipping and upending.
- There is no significant difference between the frequency of pecking and upending as well as between the frequency of dabbling and dipping.
- Northern shovelers peck and upend the least to forage, and they dabble and dip the most to forage.

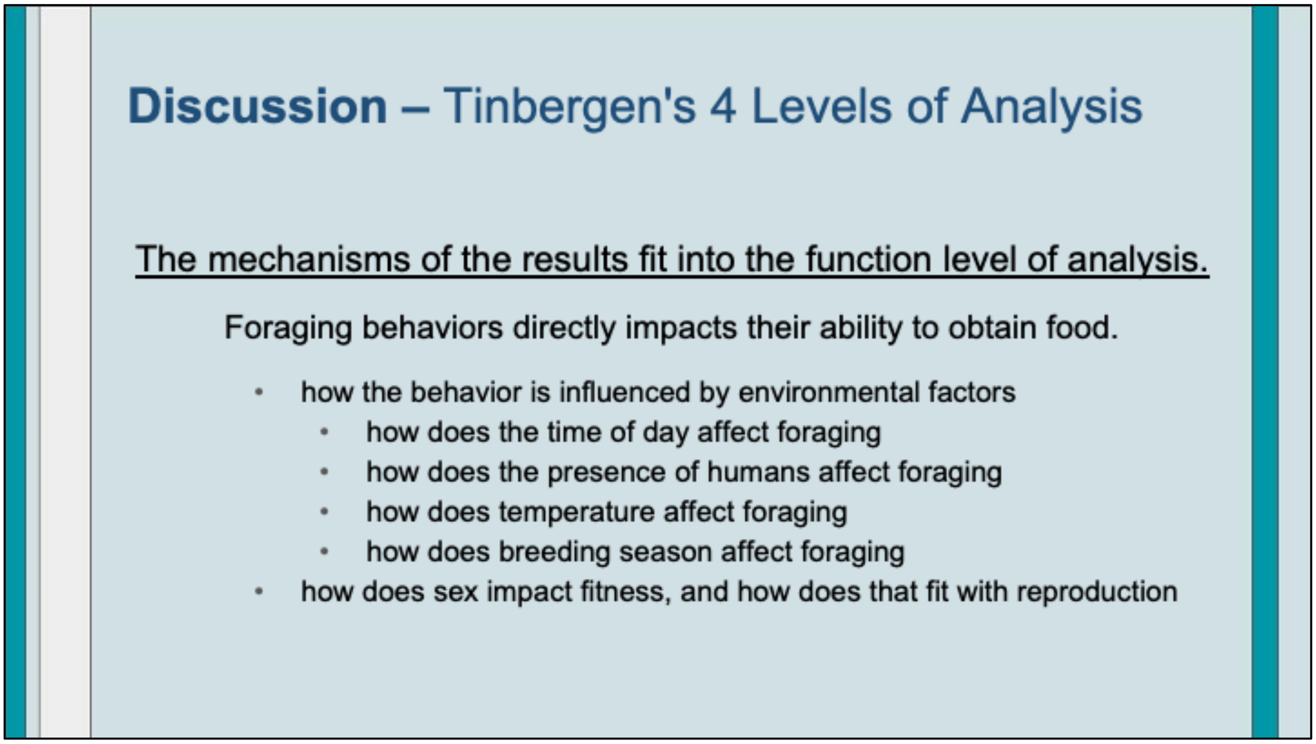
Statistical Test - Paired t-test: to determine whether the mean difference between the two sets of measurements is statistically significant or not.

- Finally, we compared the frequencies of the four foraging behaviors against each other using **paired t-tests**. The results are summarized in Table 4.
- There is a significant difference between the frequency of pecking and dabbling, the frequency of pecking and dipping, the frequency of dabbling and dipping, the frequency of dabbling and upending, and the frequency of dipping and upending. There is no significant difference between the frequency of pecking and upending as well as between the frequency of dabbling and dipping.
- The negative confidence intervals for pecking-vs-dabbling and pecking-vs-dipping show pecking was less frequent. The positive CIs for dabbling-vs-upending confirm dabbling was more frequent than upending.
- Also, **dabbling and dipping were the most frequent behaviors; pecking and upending were the least frequent**.

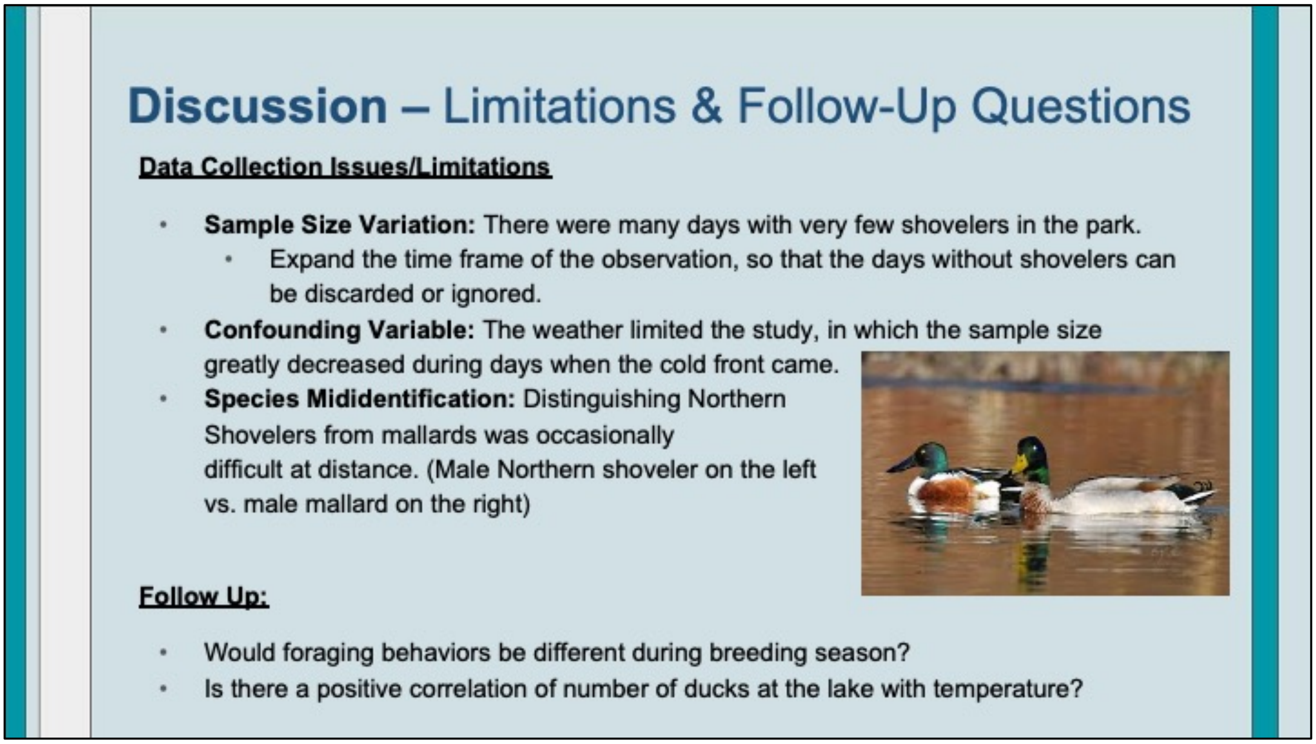


Three predictions were made prior to data collection. Two were confirmed and one was partially disproved.

- The Wilcoxon-Mann Whitney test supported the prediction that foraging frequency would not differ significantly between males and females. Even though females are known to forage more than males in the wild, the similar rates we observed align with Rose who found that captive ducks show reduced alertness around humans, which may have equalized foraging rates between sexes.
- The Kruskal-Wallis result confirmed a significant increase in foraging frequency across the day. Foraging was lowest in the morning and highest in the evening, aligning with Guillemain's conclusion that foraging activity and location shift throughout the day, partly in response to prey distribution. An additional factor observed in this study was that shovelers were predominantly asleep during morning hours — a confounding variable not anticipated in the original hypothesis, and there was an inverse relationship between human presence and foraging frequency.
- While dabbling was among the most frequent behaviors, it was statistically indistinguishable from dipping. Our original prediction was too narrow; the data support a revised conclusion that both dabbling and dipping function as the primary foraging strategies, with pecking and upending serving as secondary behaviors. However, the difference may reflect the role of human feeding of the ducks at Mueller Park, which could promote shallow-water behaviors like dipping.



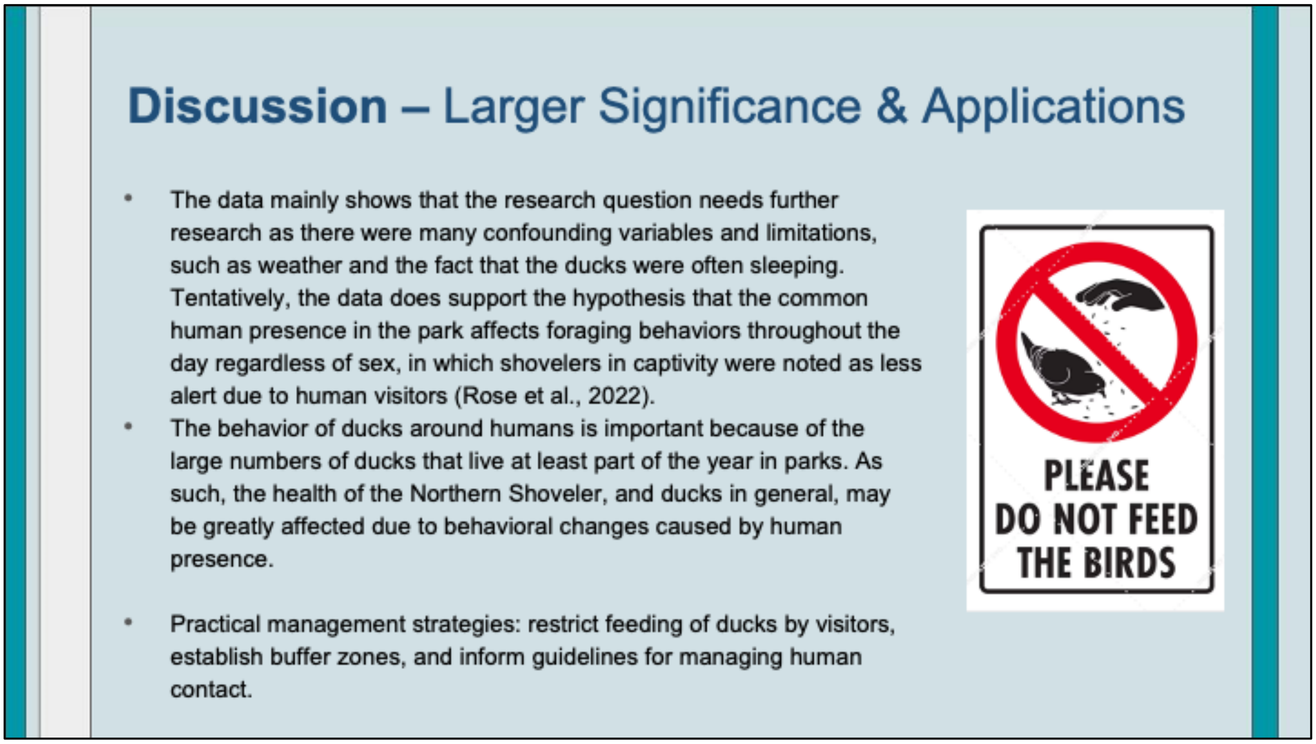
- Our findings are best interpreted at the functional level of Tinbergen's (1963) four levels of analysis, which asks what adaptive value a behavior provides. Foraging directly enables energy acquisition, which underpins survival, migration, and reproductive fitness. The time-of-day variation in foraging frequency reflects how environmental factors — including temperature, prey availability, human presence, and endogenous activity cycles — interact to shape energy acquisition strategies in a park-dwelling population.
- Future work could additionally examine the mechanistic level (hormonal or neurological triggers of foraging initiation), the developmental level (how individual shovelers learn to forage in the presence of humans), and the phylogenetic level (whether dabbling ducks as a group show similar park-mediated behavioral plasticity)



- Our study had some notable limitations. **Sample size varied dramatically** — some days had zero shovelers present, which reduced our data quality. A longer observation window would allow us to discard low-count days without losing too many usable data points.
- **Weather was a confounding variable** — cold fronts noticeably reduced duck numbers. Future studies should either control for temperature or explicitly model it as a predictor.
- **Species identification** was occasionally challenging from a distance, particularly distinguishing shovelers from mallards.

Key questions emerging from this study include:

- Would foraging behaviors differ significantly during breeding season (April–July)? Afton (1979) found dramatically higher foraging rates at the onset of breeding, and our pre-breeding observation window may have missed important behavioral variation.
- Is there a positive correlation between daily temperature and shoveler group size at the lake? Informal observations suggested fewer ducks on colder or cloudier days, but this was not formally tested.



- The bigger picture here is one of conservation relevance. Our data tentatively support the idea that human presence in public parks influences shoveler foraging behavior, even if the causality is not fully isolated. With Northern Shoveler populations declining due to habitat loss, understanding how park environments alter their behavior matters for their long-term health.
- Mueller Lake Park alone hosts over a hundred shovelers on peak days. If behavioral patterns — like reduced alertness and increased human-facilitated feeding — become the norm, it could affect their nutritional efficiency, conditioning for migration, and ultimately population fitness.

Practical management strategies based on these findings include:

- Restricting duck feeding by park visitors during critical foraging periods, as supplementary food likely alters natural foraging behavior.
- Establishing buffer zones that limit human approach distance to resting and nesting areas, particularly during morning hours when birds are inactive.
- Applying these findings to captive breeding programs to calibrate appropriate levels of human contact during rearing.

Literature Cited

Afton, A. (1979). Time Budget of Breeding Northern shovelers. The Wilson Bulletin. JSTOR, 91, 42–49. http://www.jstor.org/stable/4161165.

Guillemain, M., Fritz, H., Guillon, N. (2000). Foraging Behavior and Habitat Choice of Wintering Northern Shoveler in a Major Wintering Quarter in France. Waterbirds: The International Journal of Waterbird Biolog. JSTOR, 23, 353-363. https://doi.org/10.2307/1522173.

Rose, P., Roper, A., Banks, S., Giorgio, C., Timms, M., Vaughan, P., Hatch, S., Halpin, S., Thomas, J., O'Brien, M.(2022). Evaluation of the Time-Activity Budgets of Captive shovelers (Anatidae) Compared to Wild Counterparts. ScienceDirect, 251. https://doi.org/10.1016/j.applanim.2022.105626.

Webb, M. and Brotherson, J.D. (1988). Time-Activity Budgets of Drake Gadwall and Northern shovelers on Industrial Cooling Ponds During Late Winter and Early Spring in Central Utah: The Great Basin Naturalist. JSTOR, 48, 280-284. https://www.jstor.org/stable/41712437.

Thank you.